

EDUCATION AND COURSEWORK			
Course	College/University	Year	CGPA/%
PGD (Statistical Methods and Analytics)	Indian Statistical Institute	2024-25	7.0 CGPA
B.Tech.	National Institute of Technology, Durgapur	2017-2021	7.48 CGPA
Higher Secondary	Bijendra Public School	2016	81%
Matriculation	S.R.D.A.V Public School	2014	10 CGPA

- PROJECTS
- Hate Speech Detection Using LSTM Deep Learning Model** (*Deep Learning, tensorflow, keras, spacy, NLP*) [Github](#)
 - Designed and trained an **LSTM**-based neural network for hate speech detection on social media text, utilizing a 3-layer configuration with 100, 50, and 50 units respectively.
 - Executed comprehensive text preprocessing including **lemmatization, removal of stopwords, tokenization, padding** and **word embedding** to convert the text data into 100-dimensional vector representations.
 - Attained **88% accuracy** and **90% F1-score** on benchmark dataset
 - Credit Card Fraud Detection** (*sklearn, Pandas, Seaborn, Numpy, ML*)
 - Developed a classification system to flag fraud detection using a **100K transaction** dataset from UCI, applied Machine Learning & Deep Learning techniques.
 - Addressed **imbalanced datasets** using oversampling (**SMOTE**) and under-sampling strategies to enhance model robustness and expanded the dataset to 200k samples .
 - Reached 95% accuracy & minimized **false negatives (FN) to 4**, and XGBoost chosen for optimal performance.
 - Exploratory Analysis and Predictive Modeling of Diabetes Patient** (*sklearn, Pandas, Seaborn, NumPy, ML*) [Github](#)
 - Conducted extensive data preprocessing including addressing **multicollinearity, outlier detection**, and **missing data handling** to ensure data integrity for subsequent analysis. Performed **feature extraction** to discretize continuous variables into 5 categorical bins across 4 key columns.
 - Applied regression analysis techniques and machine learning models **including Linear Regression, Decision Tree, Random Forest**, and **XGBoost** to analyze the relationships between Features. Accuracy of XGboost and random forest came as highest accuracy of 88%.
 - Performed **Hyperparameter tuning** via **GridSearchCV** and used **cross validation** reached 89% predictive accuracy.
 - ECG Signal Classification using a Hybrid (KAN-CNN Model)**
 - Developed a hybrid deep learning model combining **Kolmogorov–Arnold Networks (KAN)** and Convolutional Neural Networks (CNN) for accurate classification of ECG signals into 5 different categories.
 - Extracted features from raw ECG signals and compared performance metrics (**accuracy, precision, recall**) of the Hybrid KAN-CNN model against standalone **KAN** and **CNN** architectures with 99% accuracy.
 - Basic EDA using SQL on E-commerce dataset** (*Using, MySql, Python*) [Github](#)
 - Utilized advanced SQL concepts like **windows functions, subqueries, joins**, and **aggregations** to conduct exploratory data analysis and answer 14 different questions based on the dataset.

- SKILLS, TOOLS & CERTIFICATIONS
- Programming Languages** : Python, Java, JavaScript, R, HTML, CSS
 - Data Science Libraries**: NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, TensorFlow, Keras, NLTK, Spacy
 - Analytics / Data Tools**: PowerBI, MySQL, Git, Excel (Intermediate)
 - Skills**: Data Analysis, Machine Learning, Deep Learning, Feature Engineering, EDA (Exploratory Data Analysis), Web Scraping, NLP
 - Concepts / Theories**: Probability & Statistics, Data Structures, Algorithms
 - Certifications** : [Python](#), [SQL\(intermediate\)](#) , [SQL\(Basic\)](#) , [Java\(Basic\)](#) ,[Problem Solving\(Basic\)](#)

- SCHOLASTIC ACHIEVEMENTS AND RESPONSIBILITY
- Solved more than 250 DSA questions across different platform - [Leetcode](#)
 - All India Rank-7 in ISI PGDSMA Entrance Exam
 - Placement Coordinator in ISI Chennai
 - Represented college in All India Inter NIT Sports tournament 2020 held at MNIT Jaipur, in Chess Competition [Certificate](#)
 - Qualified GATE CE three times. 2021,2022,2024